

# Vijay Mohanram Iyer

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🌐 [Vijay's Portfolio](#)

🌐 [Vijay Iyer](#)

## SUMMARY

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Graduate engineer specializing in computer vision, machine learning, and embedded systems for intelligent safety and health applications. Experienced in developing and fine-tuning deep learning architectures, transformer models, and end-to-end pipelines. Skilled in optimization of dynamic systems, signal processing, and deploying efficient models on embedded platforms for real-time, user-centered solutions.

## TECHNICAL SKILLS

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- **Programming Languages:** Python, C++, JavaScript, SQL, Node.js, HTML, CSS
- **Libraries/Frameworks:** OpenCV, Qt, TensorFlow, PyTorch, pandas, NumPy, Matplotlib, scikit-learn, React.js
- **ADAS:** Sensor fusion, camera and LiDAR calibration, Point cloud, object detection & tracking
- **Embedded Systems:** ESP32, STM32, Arduino, Raspberry Pi, RTOS, Embedded C/C++
- **Tools & Cloud Platforms:** Git, Docker, MATLAB, Linux, Carla, conda, LaTeX, Jupyter Notebook, Azure, AWS

## CORE COMPETENCIES

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- **Technical Leadership:** Project Development, Cross-functional Collaboration, Innovative mindset.
- **Problem Solving:** Critical Thinking, Performance Optimization, Debugging.
- **Communication:** Active Listening, Documenting, Knowledge Transfer.
- **Spoken Languages:** English C2, German A2 (Learning)

## PROFESSIONAL EXPERIENCE

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- Forschungszentrum Informatik September 2025 – Present

**Research Assistant**

  - Evaluate and optimize detection models through transfer learning for hyperparameter fine-tuning.
  - To utilize 3D geometric facemesh for reconstruction of desired regions.
  - Exploring GAN based approaches and diffusion techniques and their architectures for video rendering .
  - Conduct literature review and work towards research publication.
  - **Technology:** *Python, PyTorch, Gen AI, OpenCV*
- TecoLab March 2023 – Present

**Working Student**

  - Sensor optimization for open-earables with TinyML and quantized neural networks for on-device inference.
  - ML4Print: Designed a CNN-based pipeline to classify printing techniques and paper substrates, using saliency maps for explainability and reducing manual forensic effort by 80%.
  - Applied regression models and data integration to simulate liquid thermal behavior in industrial valves, improving predictive maintenance and system reliability.
  - **Technology:** *Python, SciPy, pandas, NumPy, Arduino, Edge ML*
- Forschungszentrum Informatik February 2025 – August 2025

**Master Thesis (Grade: 1,0)**

  - Developed an end-to-end deep learning pipeline using rPPG signals for artifact evaluation in vital signals and benchmarked under real-world conditions.
  - Designed a Fusion Model combining Vision Transformer and CNN features via a custom Fusion-Head for DeepFake classification.
  - **Technology:** *Python, PyTorch, Butterworth Filters, pandas, NumPy, Docker, Linux, conda, LaTeX*

#### 4. Access@KIT

March 2023 – May 2023

##### Working Student

- Implemented a BERT-based text-to-speech framework leveraging phoneme alignment and acoustic features to generate natural, high-quality speech.
- Refined the speech synthesis pipeline via data preprocessing and feature extraction, delivering clearer and more accessible speech for end users.
- **Technology:** *Python, PyTorch, Diffusion Models, TTS, Audio Processing, React.js, MUI*

#### 5. Accenture India Pvt. Ltd.

February 2022 – April 2022

##### Associate Software Engineer

- Maintained IBM Mainframe servers with batch job scheduling and performance monitoring.
- **Technology:** *COBOL, Batch Processing, System Monitoring*

#### 6. Accur Digitus

June 2021 – January 2022

##### Web Developer Intern

- Developed responsive web applications using React.js and Tailwind CSS, integrated scalable RESTful APIs, and enhanced user experience, resulting in over 30% increase in engagement.
- **Technology:** *React.js, Tailwind CSS, REST APIs, JavaScript, Node.js*

## EDUCATION

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### M.Sc. Electrical Engineering and Information Technology

May 2022 – September 2025

Karlsruhe Institute of Technology

GPA: 2.3

### Bachelor of Engineering in Electronics Engineering

August 2017 – May 2021

University of Mumbai, India

GPA: 2.8

## PROJECTS

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### 1. Non-Contact based Vital Extraction for Health Monitoring with Attention Mechanisms (FZI)

Utilized Plane-Orthogonal-to-skin and Wavelet Transform to extract pulsatile components from facial videos, improving robustness for non-contact vital sign monitoring in telemedicine.

**Technology:** *Python, PyTorch, OpenCV, NumPy, Matplotlib, ViT, GenAI*

### 2. CamCussion (Zeiss Innovation Hub)

Utilized pupil segmentation, gaze tracking, and saccadic velocity estimation to analyze pupil dilation and eye movements in real time, contributing to accurate concussion diagnosis.

**Technology:** *Python, OpenCV, Dlib, PyQt, Image Processing, PyTorch, NumPy, ML*

### 3. Real-Time 3D Obstacle Detection Using Raw LiDAR Point Cloud Data

Developed real-time 3D obstacle detection by voxelizing LiDAR point clouds into ViT tokens and applying multi-head attention for improved accuracy.

**Technology:** *Python, PyTorch, Open3D, NumPy, ViT, LiDAR Processing*

### 4. Real-Time Car Accident Alert System

Built ESP32-based embedded LSTM system with TensorFlow Lite for real-time crash detection and automated SMS/email alerts with GPS location.

**Technology:** *Python, ESP32, Deep Learning, Embedded C*